

BECK HERMANN

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FE-certified Mechanical Engineering M.S. student with strengths in mechanical design, CAD-driven development, and engineering analysis. Skilled in modeling components, assemblies, and systems using SolidWorks, AutoCAD, and simulation tools, supported by hands-on manufacturing and prototyping experience. Seeking a mechanical design engineering role focused on material science component design, analysis, and build-ready solutions.

EDUCATION

University of Colorado, College of Engineering and Applied Science, Boulder, CO Expected May 2027

M.S. in Mechanical Engineering, Bachelor's-Accelerated Master's Program

Pursuing Executive Engineering Leadership Certificate, Lockheed Martin Engineering Management Program

- **GPA: 3.9**

University of Colorado, College of Engineering and Applied Science, Boulder, CO May 2026

B.S. in Integrated Design Engineering (IDE) - Mechanical Engineering Emphasis, (ABET Accredited), Cum Laude

- **Major GPA: 3.768** (Overall GPA: 3.702); Dean's List: Fall '22, Spr '23, Fall '24, Spr '25, Fall '25, Spr '26
- Recipient, College of Engineering Community Impact Award
- Relevant coursework: Engineering Leadership, Engineering Ethics & Society, Engineering Projects, Computer Science II (Data Structures), Invention & Innovation, Flow Visualization, Indoor Air Pollution, Design for Inclusion

EXPERIENCE

HNTB, Denver, CO May 2024-Present

Systems Engineering Intern

- Develop RTD Light Rail system models in OpenTrack, supporting vehicle performance simulation and operational analysis for transit planning scenarios
- Contribute to strategic proposals and pursuit development for \$25M+ transportation infrastructure projects
- Conduct site visits and field documentation, capturing photography and observations to inform proposal research and design development
- Review and analyze a regional transportation visioning study and perform enterprise and agency research across CDOT and RTD, identifying strategic gaps and proposing actionable improvements to support HNTB's project pipeline development and business strategy
- Manage systems-level CAD operations using AutoCAD, MicroStation, ProjectWise, and Bluebeam, creating and modifying route and aspect sheets, automating title block and attribute updates, and maintaining xrefs, base files, and cloud-based document control to support track planning efforts
- Collaborate with engineers and PMs to ensure quality assurance, schedule adherence, and 4-for-4 deliverables
- Develop climate change adaptation strategies for extreme heat, redlining PE plans to design and cost shading solutions for escalator and elevator areas across 3 stations and researching rail coatings to prevent thermal expansion, collaborating with the client and SMEs
- Identify fiber optic communications equipment and develop rack elevation drawings across multiple light rail corridors using netTerrain and AutoCAD, establishing a baseline for future network modernization efforts
- Selected to continue part-time during the academic year, working remotely with the systems engineering team

College of Engineering and Applied Sciences, Boulder, CO

August 2023-December 2023

Course Assistant - Engineering Projects GEEN 1400

- Support professors and the IDE department to ensure smooth course operations and project execution
- Hosted office hours, guiding students through engineering concepts, design projects, and technical challenges
- Collaborated with faculty and course assistants to enhance student learning outcomes and course effectiveness

CERTIFICATIONS

- Fundamentals of Engineering (FE) Exam, Other Disciplines January 2026
- AI Fundamentals Certified April 2025
- Data Literacy Certified February 2025
- SolidWorks CSWA Certified May 2023
- OSHA 10 & CPR Certified 2021

TECHNICAL SKILLS

Engineering & Analysis: Systems Engineering, Rail Transit Simulation (OpenTrack), GIS (ArcGIS), GD&T

CAD & Design Tools: AutoCAD, MicroStation, ProjectWise, Bluebeam, SolidWorks, netTerrain (Graphical Networks)

Professional Skills: Proposal & Pursuit Strategy, Client & Business Research, Proactive Leadership & Collaboration

Programming & Technology: MATLAB, EES, C++, Arduino, AI Tools (Claude Code, Copilot, ChatGPT), LabVIEW

Fabrication and Manufacturing: 3D Printing, Laser Cutting, Lathes & Mills, Woodworking, Circuitry & Soldering

INTERESTS & EXTRACURRICULARS

- Integrated Design Engineering Student Advisory Board, Mentoring
- Skiing, Camping, Hiking National Parks, Surfing, International Travel, Woodworking, Quilting, Cooking